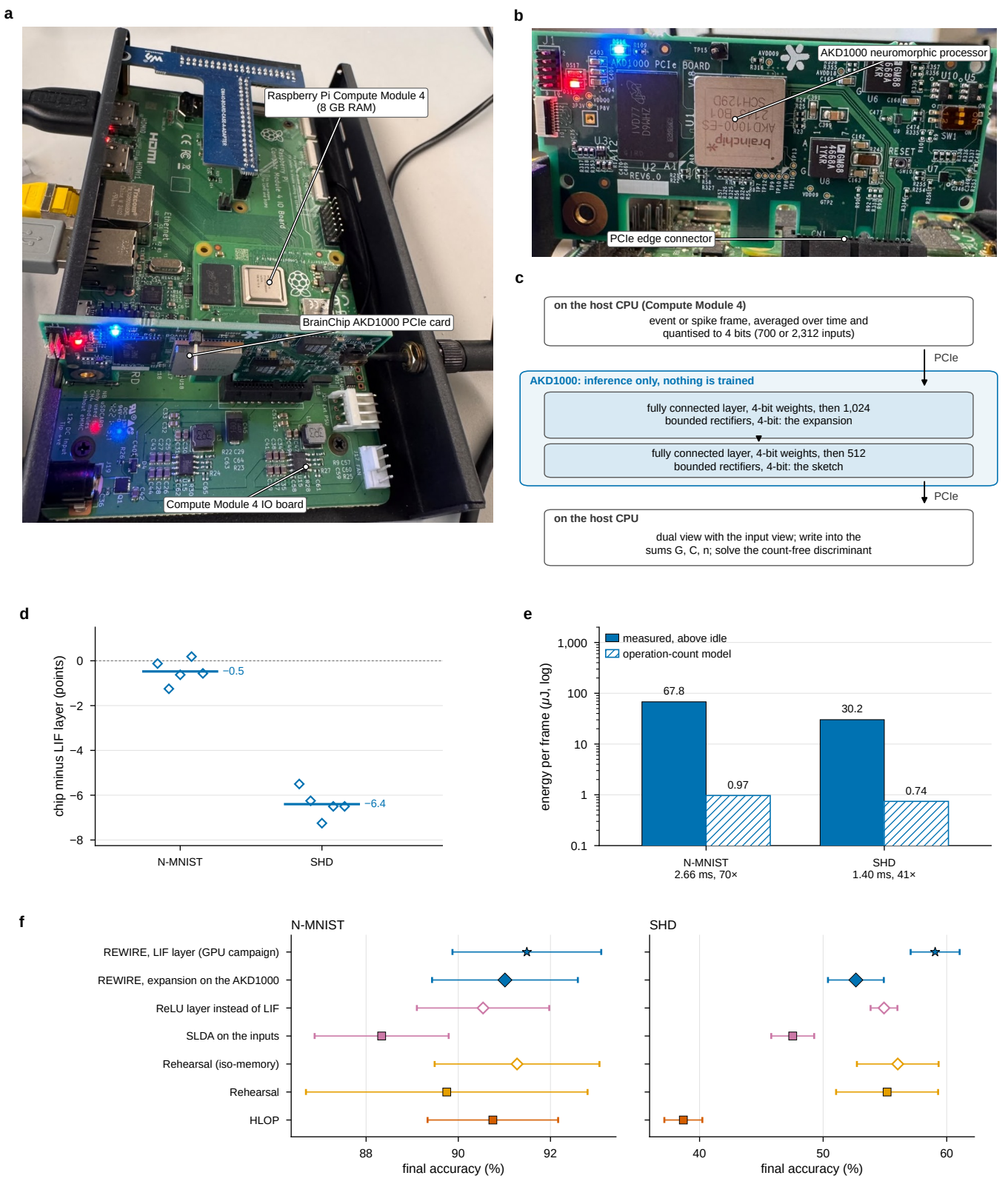

a
Raspberry Pi Compute Module 4
(8 GB RAM)
BrainChip AKD1000 PCIe card
Compute Module 4 IO board
b
AKD1000 neuromorphic processor
PCIe edge connector
c
on the host CPU (Compute Module 4)
event or spike frame, averaged over time and
quantised to 4 bits (700 or 2,312 inputs)
PCIe
AKD1000: inference only, nothing is trained
fully connected layer, 4-bit weights, then 1,024
bounded rectifiers, 4-bit: the expansion
fully connected layer, 4-bit weights, then 512
bounded rectifiers, 4-bit: the sketch
PCIe
on the host CPU
dual view with the input view; write into the
sums G, C, n; solve the count-free discriminant
d
chip minus LIF layer (points)
0
−2
−4
−6
−8
−0.5
−6.4
N-MNIST
SHD
e
energy per frame (μJ, log)
1,000
100
10
1
0.1
measured, above idle
operation-count model
67.8
0.97
30.2
0.74
N-MNIST
2.66 ms, 70×
SHD
1.40 ms, 41×
f
N-MNIST
SHD
REWIRE, LIF layer (GPU campaign)
REWIRE, expansion on the AKD1000
ReLU layer instead of LIF
SLDA on the inputs
Rehearsal (iso-memory)
Rehearsal
HLOP
88
90
92
40
50
60
final accuracy (%)
final accuracy (%)